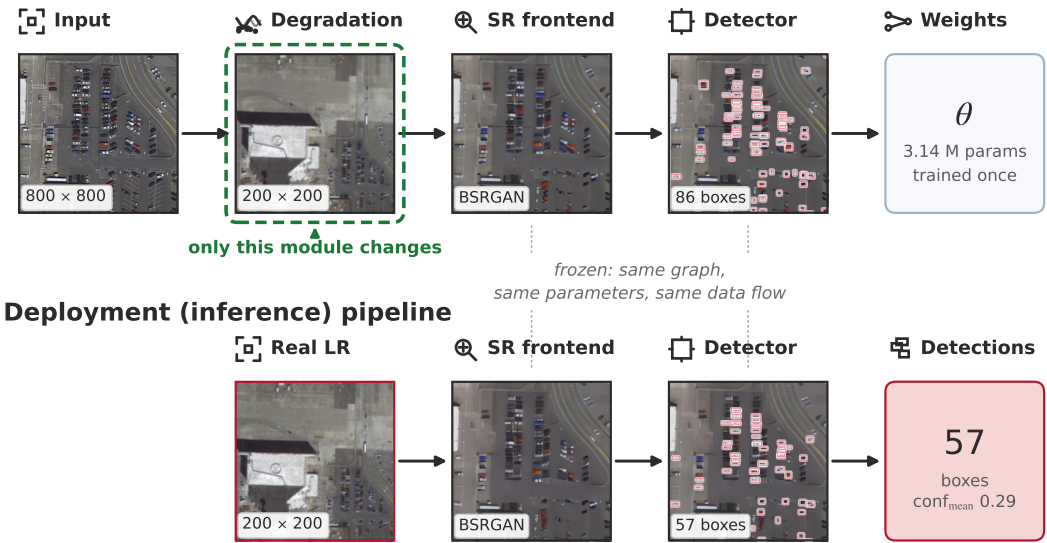

Training pipeline
Input
Degradation
SR frontend
Detector
Weights
800 × 800
200 × 200
BSRGAN
86 boxes
θ
3.14 M params
trained once
only this module changes
frozen: same graph,
same parameters, same data flow
Deployment (inference) pipeline
Real LR
SR frontend
Detector
Detections
200 × 200
BSRGAN
57 boxes
57
boxes
conf mean 0.29
Degradation switches of the four arms
A blur σ: 0
noise
JPEG
B0 blur σ: U(0, 0.05)
noise
JPEG
B2 blur σ: U(0.2, 2.0)
noise
JPEG
D0 blur σ: U(0.2, 2.0)
noise
JPEG